

Decentralized Renewable Energy for Enterprise

An inevitable next step. Join the DRE revolution.

Ten problems • Four themes • Build it, measure it, explain it



Physical prototype Hybrid Software, hidden scenario

[Event dates, venue and registration details to be added by the organizers]

Details will be shared with registered teams.

Why this is important

The global commercial landscape is undergoing a silent structural shift. For decades, micro, small, and medium enterprises (MSMEs) and large-scale industries alike have relied on centralized grids to fuel their daily operational needs. This top-down dependency brings real vulnerabilities: grid instability, thermal overloads, fluctuating commercial tariffs, and a heavy carbon footprint.

35% possible increase in annual business income	0.5 M direct jobs globally, plus 1.7 M indirect	49% of global installed power capacity is renewable	<6% of Andhra Pradesh's 167 GW potential is in use
---	---	---	---

Why should enterprises shift to DRE?

- Substantial improvements measured directly in **operational continuity and baseline productivity**.
- Enables up to a **35% increase in annual business income**, bridging the gap where centralized grid power remains expensive or unreliable.
- Generates close to **half a million direct jobs globally** (and 1.7 million indirect jobs), matching the traditional footprint of localized oil and gas workforces.

Today, **Decentralized Renewable Energy (DRE)** solutions, from rooftop solar installations and local microgrids to IoT-enabled productive-use appliances, are moving from backup options to primary drivers of corporate productivity and operational resilience. The world is moving quickly towards clean infrastructure. Renewables account for 49% of all global installed power capacity, and almost all of the net additions (96.8%) came from variable solar and wind systems. Decentralization is no longer a future concept; it is a current reality.

What is happening in our region?

India installed 55.3 GW of non-fossil capacity additions in the last fiscal year and now stands third globally in installed renewable capacity. Andhra Pradesh is among the top states in India, holding 9.1% of the nation's renewable energy generation potential. Despite the massive potential (up to 167 GW), less than 6% of this capacity is being used.

The state's policy identifies wind and pumped hydro storage as the two main sources of renewable energy in the north coastal districts, in addition to solar. While the government pushes to meet its policy goals, the ground reality of our region throws up challenges that need innovative solutions:

- **Uncoordinated rooftop solar.** Rapid commercial adoption in dense corridors like Gajuwaka or Atchutapuram pushes heavy, volatile reverse power back into local distribution networks on sunny afternoons, causing voltage spikes that stress substation transformers.
- **Hills, forests and unreliable power.** Hilly topography and dense forest cover make expanding the central grid physically and financially prohibitive. Tribal processing cooperatives (coffee, pepper and millets in Paderu or Chintapalli) hesitate to invest in processing capacity because power is unreliable or expensive. Isolated DRE machinery suits them well, but when an inverter or motor faults, the lack of local diagnostic tools and skilled technicians means weeks of downtime and spoiled harvests.
- **Surge currents in aquaculture.** Agricultural and shrimp-farming enterprises need continuous energy for cold storage and pond aerators. Heavy inductive machinery draws large inrush currents that trip low-cost rural

solar microgrids and accelerate battery degradation, making solar-plus-storage financially difficult for local cooperatives.

These create specific challenges for enterprises. The next pages set out **10 problem statements in four themes. Pick one and provide a solution.**

How this hackathon works

Energy problems are physical problems. Heat, moisture, motors and batteries behave the way physics says they do, not the way a dashboard says they do. So this hackathon rewards teams that **build, measure and explain**, not teams that only present.

Three kinds of problems

PS 1, 2, 4, 8	Physical prototype required. Bring a working, bench-scale prototype and run it live. A slide deck or simulation cannot replace it.
PS 5	Hybrid. A software system plus a working physical component (a real lock mechanism).
PS 3, 6, 7, 9, 10	Software on a hidden scenario. Your tool is run live on a scenario or dataset that the organizers reveal only on demo day.

Using AI tools

AI tools are allowed. They are useful for routine work such as debugging or drafting code. What they cannot do is build and tune your hardware, learn what a jaggery maker or a hatchery operator actually needs, or answer the judges' questions for you. Marks are therefore weighted toward the live demo, your reasoning and your field understanding. Every team must:

- 1 Keep a design notebook** (paper or digital) with sketches, hand calculations, assumptions, decisions and failed attempts. Bring it to the demo.
- 2 Speak to at least one real user or expert** (a processor, farmer, operator, technician or engineer) and submit five lines on what you learned and what you changed because of it.
- 3 Submit a one-page AI-use log:** which tools you used, for what, and at least one case where the AI's output was wrong or unsuitable and how you corrected it. Teams that catch and fix AI errors earn bonus marks.
- 4 Respect the constraint sheet:** component budget cap of [₹5,000, to be confirmed]; the demo runs without internet; the solution must be usable by an operator with basic literacy and no technical training; prefer locally available materials.
- 5 Expect a surprise test and an oral defence.** On demo day the judges will change a condition without warning, and will question the whole team. Any member may be asked to explain any part of the solution.

SAFETY

No open flames at the venue. Keep prototypes at low voltage wherever possible. Any mains-powered heater or motor needs a fuse or breaker, a thermal cut-off where relevant, and a mentor sign-off before it is switched on.

A

THEME A

Agriculture & Horticulture Value Chains

01

Solar-Assisted Jaggery Processing

Closed-Loop Heat Control Rig

PHYSICAL PROTOTYPE REQUIRED

VALUE CHAIN Sugarcane and jaggery.**The problem**

Local micro-enterprises struggle with inconsistent quality during sugarcane juice boiling because biomass or solar-hybrid heat is distributed unevenly. Overheating or underheating ruins the jaggery grade.

Your task

Build a bench-scale controller that keeps a boiling vessel on a target temperature profile even when the heat source varies. A hot plate or heating element can stand in for the biomass and solar-hybrid heat.

Deliverables

- A working rig: a vessel (water or sugar syrup), at least two temperature sensors at different positions, and at least one actuator (damper, valve, heater dimmer or relay) under closed-loop control.
- A predictive “pour now” alert (light, buzzer or display) based on the measured rate of temperature change, not a fixed timer.
- An automatic batch record that grades the batch from temperature consistency, viewable on a phone or a simple display.
- A bill of materials with total cost, and a short note on how the design would scale to a real jaggery pan.

LIVE TEST ON DEMO DAY

Standard test. Follow a scaled-down, safe temperature profile issued on the day, within the tolerance announced with it.

Surprise test. Judges disturb the system (change the heat input, add cooler liquid, or open the lid) and the controller must recover.

Show your thinking

- Estimate by hand the heat needed to raise and hold the batch temperature and to evaporate a stated quantity of water. Compare it with what your rig measured.
- Explain which control method you chose (on/off, proportional, PID or other) and why.

02 Solar Dryer Moisture Tracker

Weight-Based Drying Monitor

PHYSICAL PROTOTYPE REQUIRED

VALUE CHAIN Coffee cultivation and cocoa beans.

The problem

Most farmers dry their produce under open sun, at the mercy of changing weather. Solar tunnel dryers or large collective covered drying facilities help, but those using tunnel dryers still guess when the beans are properly dried. Taking them out too early causes mold; leaving them in too long reduces weight and profits.

Your task

Build a small instrumented dryer (enclosure or drying tray) that estimates the moisture of a sample and predicts the remaining drying time.

Deliverables

- A dryer enclosure or tray with a fan or vents, temperature and humidity sensors, and a load cell that tracks the sample's weight.
- A moisture estimate computed from weight loss and the starting moisture, with a live remaining-time estimate that updates as conditions change.
- An on-device alert (buzzer, light or display) when the target moisture is reached, for example 7% for coffee. SMS or WhatsApp alerts are a bonus, but the device must work without a network.
- An exportable drying log suitable for quality certification. Weather APIs may be used to improve predictions, but the device must still work when they are unavailable.

LIVE TEST ON DEMO DAY

Standard test. Organizers supply a pre-weighed damp sample with its starting moisture. Your device must call the moment the target moisture is reached. Judges verify with a reference scale or moisture meter.

Surprise test. Judges change the airflow or heat (cover the vents, switch off the heat source) and your remaining-time estimate must adapt.

Show your thinking

- Show the drying curve from your own measurements and explain why drying slows down as the sample dries.
- State the accuracy you achieved and what limits it.

03 DRE-Powered Jackfruit & Banana Flower Processing

Solar-Aware Scheduler

SOFTWARE: JUDGED ON A HIDDEN SCENARIO

VALUE CHAIN Secondary produce processing.

The problem

Tribal women's cooperatives could use solar-powered cutting, slicing and dehydration machinery. Processing must happen immediately after harvest, but variable weather can drain the solar microgrid batteries mid-operation.

Your task

Build a production scheduling and load-balancing tool for a decentralized food processing hub.

Deliverables

- A scheduler that takes harvest volumes, machine power ratings, battery capacity and a solar forecast, and recommends the best hours to run heavy slicing machines versus low-power dehydrators while avoiding battery depletion.
- A one-page printed schedule that a cooperative member can follow without a smartphone.
- A plain-language explanation of why each machine runs when it does.

LIVE TEST ON DEMO DAY

Standard test. Organizers issue a three-day scenario on demo day (harvest volumes, forecast, battery size, machine list). Your scheduler is scored on batches completed without battery depletion.

Surprise test. Judges change the forecast (a cloudy afternoon) or take a machine offline mid-plan, and your tool must re-plan live.

Show your thinking

- State in writing what you are optimizing and which limits are hard constraints.
- Explain why you chose your method (rules, optimization or heuristic) over the alternatives.

B

THEME B

Aquaculture & Cold Chain Resilience

04

Shrimp Hatcheries & Cold Storage

Thermal Pre-Cooling and Aerator Soft-Start Rig

PHYSICAL PROTOTYPE REQUIRED

VALUE CHAIN Shrimp and prawn farming.

The problem

Shifting rural cold rooms and hatchery aerators completely to solar-plus-storage mini-grids is expensive because of high battery costs. Hatcheries face immediate crop spoilage if the power drops abruptly.

Your task

Build a small-scale rig that demonstrates two ideas: store cold while the sun is available so less energy is drawn at night, and start motors gently so surge currents do not trip the system or wear the battery.

Deliverables

- An insulated mini cold box with thermal mass (water bottles or ice packs), temperature sensors, and a controller that safely over-cools during a “solar window” (a bench supply or small panel running a compressed 10 to 15 minute “day”).
- A small motor or pump (aerator stand-in) with a soft-start or ramp circuit and a current sensor that records the start-up surge with and without it.
- An emergency low-battery protocol that protects the aerator first and reduces the cold box load, demonstrated on a simulated battery state of charge.
- A measured comparison of energy drawn during the “night” phase with and without pre-cooling.

LIVE TEST ON DEMO DAY

Standard test. Run the compressed day and night cycle while judges read energy and current from an independent meter.

Surprise test. Judges cut the “sun” early or add a second load, and the protocol must keep the aerator running.

Show your thinking

- Calculate by hand the thermal energy stored (mass, specific heat, temperature change) and compare it with what you measured.
- State the safe temperature band for the stored product and how it limits how far you can over-cool. A Peltier module is acceptable as a stand-in but is inefficient, so explain how your results would scale to a real compressor system.

05 Shared Solar Cooling Hub

Pay-As-You-Go Locker with a Real Lock

HYBRID: PHYSICAL COMPONENT + SOFTWARE

VALUE CHAIN Marine fisheries and fresh vegetables.

The problem

Smallholder marginal farmers and fishermen cannot afford their own solar refrigerators. Shared community cooling hubs exist, but managing manual slot bookings and collecting payments transparently can be chaotic.

Your task

Build a booking and micro-billing system for shared solar cooling lockers, including a physical locker mechanism that only opens for a valid booking.

Deliverables

- A working locker prototype (solenoid or servo latch with a controller) that opens only after a valid booking token is presented by QR code or RFID.
- A mock UPI payment flow (no real money) that bills per hour and per kg. A load cell for weight is a bonus.
- Offline tolerance: if connectivity drops, bookings and payments are queued and reconciled later without double charging.
- A dashboard showing diesel emissions avoided by the village, with the formula and the source of the emission factor stated.

LIVE TEST ON DEMO DAY

Standard test. Present valid, expired and double-booked tokens to the physical lock. Cut the network mid-transaction.

Surprise test. Organizers issue a set of billing edge cases on the day (partial hours, early release, overweight loads) and score the amounts your system charges.

Show your thinking

- Describe a fisherman's actual day at the hub: who books, when, and how a user without a smartphone still gets access.
- Show how the emissions figure is calculated and where each number comes from.

C

THEME C

Urban Infrastructure & Industrial MSMEs

06

Commercial Rooftop Solar Clusters

Transformer Hosting Capacity Simulator

SOFTWARE: JUDGED ON A HIDDEN SCENARIO

VALUE CHAIN Industrial MSMEs, pharma clusters and large commercial establishments.

The problem

Rapid, uncoordinated rooftop solar adoption by commercial buildings pushes heavy reverse power back into local distribution transformers during sunny afternoons, risking grid failure.

Your task

Build a geospatial solar penetration and transformer hosting capacity simulator for local utility networks.

Deliverables

- An interactive map (Streamlit, Leaflet or similar) showing local transformers, where a user can drop a pin to simulate adding a new rooftop system.
- A “Grid Safety Impact Score” and a capacity threshold warning, with the method written out: which criterion you use (for example reverse power as a share of transformer rating, or voltage rise) and why.
- A clear statement of what data a real utility would need to make the tool reliable.

LIVE TEST ON DEMO DAY

Standard test. Organizers supply anonymized data for one distribution cluster and a set of test cases with reference answers prepared by a power-systems engineer. Scored on agreement with the reference and sensible warnings.

Surprise test. Judges add a second system near the first, so your tool must account for cumulative impact.

Show your thinking

- Explain where your simplifications could mislead a utility engineer.
- Show a hand calculation for one transformer that matches the tool’s output.

07 Manufacturing Clusters

Solar-Aware Demand Shifting Scheduler

SOFTWARE: JUDGED ON A HIDDEN SCENARIO

VALUE CHAIN Heavy fabrication, engineering and apparel manufacturing.

The problem

Industrial units run heavy machinery during peak grid hours, driving up power costs, while their rooftop solar arrays generate excess power that goes to waste when the factories are winding down operations.

Your task

Build a demand-shifting scheduler that matches production schedules with solar availability and time-of-use (ToU) tariffs.

Deliverables

- A scheduling tool where factory managers enter order deadlines, machinery power ratings, ToU tariff slabs and a solar profile.
- An optimized shift schedule that maximizes direct solar consumption and minimizes peak-hour grid consumption, with the cost and solar share compared against a naive schedule.
- A list of the real-world constraints you handled (machine order, changeovers, operator shifts, maximum demand charges) and those you did not.

LIVE TEST ON DEMO DAY

Standard test. Organizers issue a hidden factory case on demo day. Scored on schedule cost, deadlines met and solar share.

Surprise test. Judges move a deadline or take a machine offline, and the tool must re-plan live.

Show your thinking

- Argue how close your schedule is to the best possible and how you know.
- Explain how a factory manager would learn to trust the schedule.

THEME D

D

Tribal Livelihoods, Eco-Tourism and Field Operations

08

Tribal Machinery

Offline Fault Diagnostics on a Fault-Injectable Rig

PHYSICAL PROTOTYPE REQUIRED

VALUE CHAIN Non-timber forest produce (pepper thrashers, solar cleaners, millet hullers in remote Paderu and Chintapalli).

The problem

Processing machinery operates in remote valleys with little or no internet connectivity. When an inverter or motor acts up, local operators cannot diagnose it, and downtime drags on.

Your task

Build a small machine rig and an offline diagnostic reader that identifies faults and guides an operator through simple repairs.

Deliverables

- A rig with a motor and driver (or inverter stand-in), sensors for current, voltage, temperature and speed, and at least four faults that can be physically introduced (for example a stalled load, low supply voltage, over-temperature, a disconnected wire).
- An offline diagnostic device or app (phone via Bluetooth or local Wi-Fi, or a handheld with a display) that reads fault codes with no internet and walks the operator through simple repair steps using icons or Telugu prompts.
- A log compressed to fit a single SMS (160 characters) holding the device ID, fault, time and key readings, ready to send when a cell tower is found. Sending a real SMS is optional.

LIVE TEST ON DEMO DAY

Standard test. Judges introduce a fault while the team looks away. The device must identify it and give the correct first step.

Surprise test. Judges introduce a fault the team did not plan for. The device should say it is unknown, not give a wrong diagnosis.

Show your thinking

- Explain how each fault shows up in the sensor readings and why your readings can tell them apart.
- List what a first-time operator can safely fix and what must wait for a technician.

09 Eco-Tourism Homestays

Microgrid Sizing Planner

SOFTWARE: JUDGED ON A HIDDEN SCENARIO

VALUE CHAIN Experiential tribal homestays and eco-resorts.

The problem

Homestay owners want to go 100% green using solar and micro-hydro, but they struggle to work out how many panels or batteries they need without buying overly expensive commercial configurations.

Your task

Create an eco-resort microgrid planner and green metric tracker.

Deliverables

- A simple, multilingual (Telugu and English) web form where the owner enters room count and appliances and receives a DRE blueprint (panels, battery, inverter and, where relevant, micro-hydro) with a cost range.
- Visible, editable assumptions (sun hours, battery depth of discharge, losses, days of autonomy).
- A guest-facing display of green metrics, such as “Your stay today ran on 100% solar power, saving 12 kg of CO₂.” Every figure must come from a stated calculation and source. A low-cost physical display is a bonus.

LIVE TEST ON DEMO DAY

Standard test. Organizers issue three homestay cases with awkward constraints (a partly shaded roof, seasonal occupancy, a small stream with low head, diesel backup). An engineer reviews each blueprint for over- or under-sizing.

Surprise test. Judges add rooms or appliances and the blueprint must update.

Show your thinking

- Work through the sizing of one case by hand (daily energy, peak sun hours, battery autonomy, depth of discharge, losses).
- Explain what an owner should do when the plan turns out to be too expensive.

10 Rural Enterprise Mobilization

Voice-Based Energy Audit Assistant

SOFTWARE: JUDGED ON A HIDDEN SCENARIO

VALUE CHAIN Rural enterprise mobilization and NGO field operations.

The problem

Field coordinators conducting energy audits across hundreds of tribal villages spend hours filling out forms by hand to check whether a solar pump or food processor is viable for a given community cluster.

Your task

Build a voice-based audit and baseline tool that works in regional languages. AI or NLP tools may be used but are not required; a guided voice question flow is acceptable. What is assessed is the audit logic and how well the tool handles real speech.

Deliverables

- Voice input in Telugu and English that captures village data such as the number of active shops, current diesel costs and daily operating hours.
- Detection of missing or unclear values, with a follow-up question instead of a guess.
- A technical and economic feasibility summary in which every number traces to an input and a visible formula. No unexplained figures.
- Support for poor connectivity: record now, process later.

LIVE TEST ON DEMO DAY

Standard test. Organizers supply recorded field-style clips (Telugu and English, noisy, with ambiguous numbers and units and self-corrections). Scored on extraction accuracy and handling of ambiguity.

Surprise test. Judges speak a new clip with a mid-sentence correction (“no, 8 hours, not 6”).

Show your thinking

- Before demo day, test with at least one Telugu speaker outside your team and show what failed and what you fixed.
- Explain which parts of a feasibility summary need human or engineering judgment and should not be automated.

Problem to bottleneck mapping

Each problem targets a specific bottleneck in the DRE value chain. This table shows what it is and where to focus.

Problem statement	Type	Primary global bottleneck	Core technical focus
PS 1 Solar-Biomass Heat Control Rig	Physical	Industrial heat decarbonization gap	Closed-loop control of variable heat sources using dampers, valves or heater control, measured against a temperature profile.
PS 2 Solar Dryer Moisture Tracker	Physical	Agricultural metrology and quality risk	Weight-based moisture estimation and drying-curve prediction, checked against a reference.
PS 3 Processing Scheduler	Software	Production and intermittency mismatch	Load balancing that matches machinery run times to solar generation and battery limits.
PS 4 Thermal Storage and Aerator Soft-Start	Physical	Chemical battery capacity and lifetime cost	Soft-starting high-inrush induction motors and shifting load using thermal mass, measured on a real rig.
PS 5 PAYG Shared Cooling Hub	Hybrid	Upfront capital expense (CapEx) barrier	Physical locker access, offline-tolerant micro-billing and slot reservation (CapEx to OpEx).
PS 6 Rooftop Solar Grid Safety Tool	Software	One-way distribution grid volatility	Geospatial transformer hosting capacity simulation and reverse-power warnings, checked against reference cases.
PS 7 Solar-Aware ToU Scheduler	Software	Demand-side flexibility deficit	Time-of-use scheduling that absorbs solar abundance under real production constraints.
PS 8 Offline Machinery Diagnostics	Physical	“Last-mile” diagnostic and maintenance gap	Offline fault detection on a real rig, guided repair and SMS-sized logging.
PS 9 Homestay Sizing Planner	Software	Technical sizing and planning illiteracy	Converting appliance inventories into a verifiable clean-energy blueprint.
PS 10 Voice-Based Energy Audit Assistant	Software	Feasibility study and energy audit delays	Voice-to-structured-data extraction with transparent feasibility calculations in regional languages.

Judging

All teams are scored out of 100 on the same four criteria. For software problems, “live demo” means running your tool on the hidden scenario, followed by the surprise change.

Criterion and weight	Points	What strong looks like
A. Live performance 40 points	Meets the standard test: 20 Handles the surprise test: 15 Safety and build quality (or robustness, for software): 5	Works as specified in front of the judges, recovers or adapts when conditions change, and fails honestly (says “unknown” rather than guessing).
B. Reasoning and oral defence 25 points	Hand calculations and assumptions are correct and documented: 10 Each team member can explain the design when asked: 10 Trade-offs and rejected alternatives: 5	Numbers in the notebook match what the rig or tool does. Answers come from understanding, not from reading notes. The team can say why it did not choose another approach.
C. Field validation and user fit 20 points	Real user or expert contact, and design changes because of it: 10 Usable by the target operator (offline, language, low literacy): 10	The team learned something that changed the design, and the solution suits a real operator in Paderu, a hatchery or a homestay, not a developer.
D. Documentation, cost and scalability 15 points	Bill of materials or cost estimate and a scale-up note: 8 AI-use log and design notebook completeness: 4 Clarity of documentation: 3	Costs are realistic, the path from prototype to real use is credible, and the AI-use log is candid.
Bonus up to +5	Caught and corrected an AI error, and can show how.	Documented in the AI-use log and confirmed in the oral defence.

Demo day format

Stage	Time	What happens
1. Setup and safety check	10 min	A mentor checks wiring, power sources and any heat source. No open flames.
2. Standard test	5 min	The team runs the announced test: the prototype for physical problems, or the hidden scenario for software problems.
3. Surprise test	3 to 5 min	Judges change a condition without warning. The team adapts, or explains what would have to change.
4. Oral defence	8 to 10 min	Questions to the whole team. Judges choose who answers, and every member should be able to explain any part of the solution.
5. Scoring	After each team	Judges score independently against the rubric, then discuss any large differences.

The judging panel includes at least one electrical or electronics expert, one domain expert (agri-processing, aquaculture or rural energy) and one software or data expert.